TIC 434331752: a 1.385-day variable star

Oct 11, 2026 · @Steve Price

Summary

TIC 434331752 (Gaia DR3 3370333679817531648, V = 13.6) varies with a period of **1.385 days**. TESS (2021, 2023) and ASAS-SN (2014–2018) recover that period independently, and ATLAS's catalogued periods (2015–2017) are consistent with it: twice the period, and its daily alias.

- **Amplitude:** a folded half-amplitude of about 1% in 2021, rising to about 2.5% in 2023. The light-curve shape and minima timing change between sectors.
- **Not an instrumental artefact:** in TESS images the signal is centred on this star in every sector. Similar stars in the same images don't show it. The ground-based surveys aren't affected by TESS's 13.7-day orbit.
- **Supporting evidence:** Gaia and ZTF both show scatter several times that of comparable nearby stars.
- **Catalogue status:** no variable-star catalogue lists it as a confirmed variable. ATLAS lists it as a candidate classified `dubious`. It is also the optical counterpart of a ROSAT X-ray source, and a candidate member of a young (about 135 Myr) stellar group.
- **Likely nature:** spots on a young, magnetically active star. A second period of 2.28 days appears in 4 of 5 TESS sectors, also from this star's position. Together with Gaia's hint of a companion (RUWE 1.67), that suggests an unresolved binary in which both stars are spotted. Confirming it needs spectroscopy.

VSX submission

Submitted to VSX on Oct 11, 2026 as a new variable. VSX assigned the provisional designation **ATO J093.6734+17.5725** while the submission is checked.

Field	Value prepared
Type	BY (spotted star)

Field	Value prepared
Period	1.38477 d (ASAS-SN). The three surveys span 1.38458–1.3849 d, so about ±0.0003 d is a realistic uncertainty
Epoch of minimum	HJD 2457423.5891 (ASAS-SN)
Range	V 13.53–13.57 (ASAS-SN, 2014–2018; the TIC's V = 13.58 comes from a different calibration)
Supporting files	ASAS-SN V phase plot and light curve (133 points: 134 less one 4σ outlier); TESS phase plot by sector, 2021 and 2023

The provisional designation is the star's ATLAS catalogue identifier. Any final VSX name or type will be added here once the review is complete.

The star

A G-type star about 600 pc away that appears slightly evolved (log g 4.06, R 1.48 R☉), though an unresolved binary would mimic this (see Nature of the variability).

Property	Value	Source
TIC	434331752	TIC v8.2
Gaia DR3	3370333679817531648	Gaia DR3
ATLAS	ATO J093.6734+17.5725	Heinze et al. 2018
RA, Dec (J2000)	06h 14m 41.64s, +17° 34′ 21.3″ (93.67348°, +17.57258°)	TIC
Magnitudes	V 13.58, G 13.35, T 12.81	TIC, Gaia
Gaia colour BP−RP	1.02	Gaia DR3
Teff	5580 K (TIC); 5725 K (Gaia)	TIC, Gaia DR3
log g	4.06	Gaia DR3

Property	Value	Source
Radius	1.48 R☉	TIC
Parallax	1.67 mas (about 600 pc)	Gaia DR3
RUWE	1.67	Gaia DR3
Radial velocity	24.5 km/s (mean)	Gaia DR3

RUWE is above the 1.4 level often used to flag a possible unresolved companion. Gaia reports no variability flag and no epoch photometry for this source.

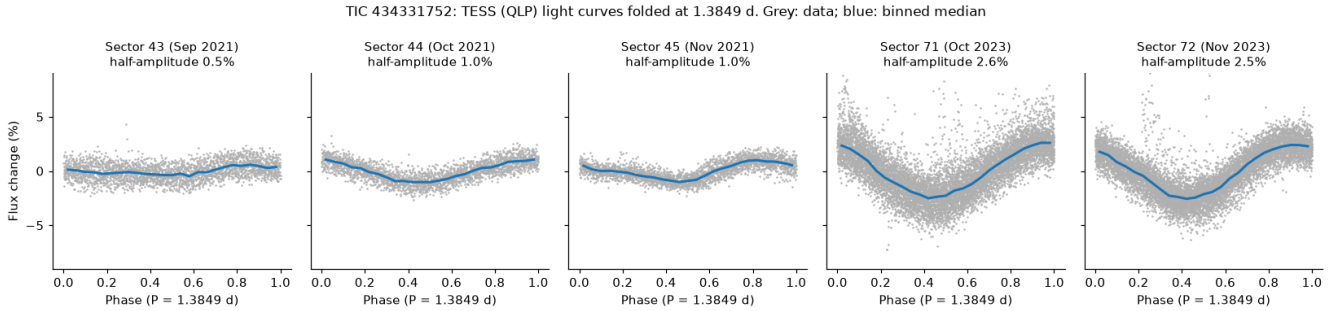
How it was found

The variability turned up as a false positive in a transit search of a neighbouring star, TIC 434331772 (T = 13.0), which is 30″ away (1.4 TESS pixels).

1. **Search:** the transit search found a 12.56-day signal on TIC 434331772. It was flagged for odd/even depth differences, a duration too long for the star, and depths varying between sectors.
2. **Pixel check:** in difference images from 4 sectors, the dimming was centred 30″ from TIC 434331772 (sector scatter 4″), 3″ from TIC 434331752. The dip was twice as deep in a 3×3-pixel aperture as in TIC 434331772's own pixel.
3. **Source light curve:** TIC 434331752's own TESS light curve is dominated by a 1.385-day variation. Once that is removed, it shows no dip at the candidate's times (0.3σ). So the 12.56-day "signal" was a chance alignment of this star's minima leaking into its neighbour's aperture, not an eclipse.

TESS light curve

In four of the five TESS sectors the strongest period is 1.375–1.381 days. The amplitude rose from about 1% in 2021 to about 2.5% in 2023.

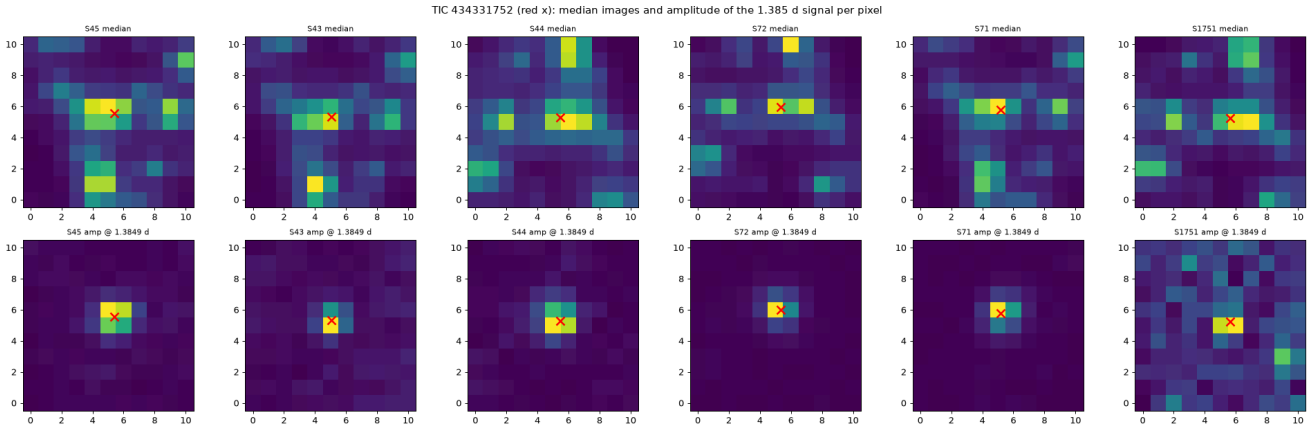


Sector	Dates	Cadence	Strongest period (d)	Second period (d)
72	Nov 2023	200 s	1.381	2.286
71	Oct 2023	200 s	1.380	2.280
45	Nov 2021	600 s	1.375	none
44	Oct 2021	600 s	1.380	2.299
43	Sep 2021	600 s	2.278	(1.385 weak)

- **Data:** QLP light curves from full-frame images. One day is trimmed from each end of every data segment to remove scattered-light ramps.
- **Combined period:** 1.3849 d from all sectors together. A single sector resolves the period only to about ±0.07 d, so the per-sector values agree.
- **Second period:** found after subtracting the 1.385-day signal. In sector 43 it is the stronger of the two.
- **Shape changes:** the minimum is broad and the shape differs from sector to sector, as expected when spots grow, decay and move.
- **January 2026:** a 7-day observation (labelled sector 1751) is too short to confirm the period after trimming. Its scatter (0.18%) suggests a lower amplitude.

Ruling out TESS systematics

The signal comes from this star, not the spacecraft. Three tests were run on 11×11-pixel cutouts of the full-frame images.



1. **Per-pixel amplitude maps** (bottom row above): a 1.385-day wave was fitted to every pixel. In sectors 44, 45, 71 and 72 the amplitude centres 1–6″ from TIC 434331752, with 51–68% of the total in the 3×3 pixels around it. Sector 43 is also centred on the star. Pointing jitter or scattered light would spread across the frame instead.
2. **Neighbouring stars:** other stars of similar brightness in the same cutouts (T = 12.8–13.6, 77–115″ away) show Lomb–Scargle power of at most 0.11 at 1.385 d, against 0.63–0.76 for TIC 434331752. The one exception is TIC 434331772, 30″ away, whose aperture overlaps; that leakage is how the signal was first found.
3. **The TESS orbit:** TESS orbits every 13.7 days, and the orbit's 10th harmonic, 1.37 d, can't be told apart from 1.385 d within one sector. The ground-based surveys below aren't tied to TESS's orbit, and they recover 1.3846–1.3848 d.

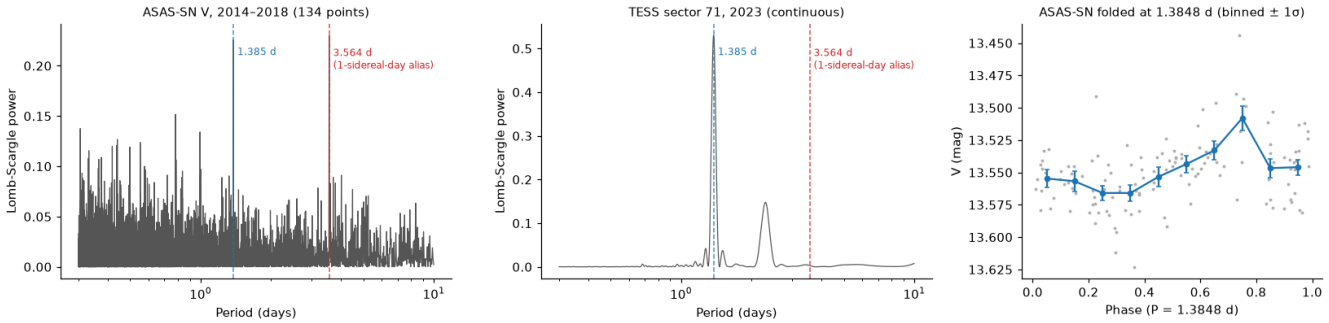
The 2.28-day signal was mapped the same way in sectors 43, 44, 71 and 72. Its centre lies 4–10″ from TIC 434331752. The only other Gaia source within 20″ is 7 magnitudes fainter (G = 20.3 at 18″), far too faint to produce it.

Independent confirmation

Two ground-based surveys recover the same period years before TESS. Two more show the star is unusually variable.

Survey	Years	Data	Result
ASAS-SN	Dec 2014 – Mar 2018	134 V-band points (133 after removing one 4σ outlier)	Peak at 1.3848 d (false-alarm probability 8×10^{-4}), with its alias at 3.564 d

Survey	Years	Data	Result
ATLAS	2015–2017	336 points (c and o bands)	Lomb–Scargle period (ls-Pday) 2.769157 d = 2 × 1.38458 d; Fourier "dominant period" (fp-domperiod) 3.5635 d, the daily alias; 2.3% rms
Gaia DR3	2014–2017	213 G-band observations (no epoch photometry published)	Variability proxy 1.5%, above 90.6% of 171 similar stars within 30′ (median 0.40%)
ZTF	2018–2025	109 r, 61 g points	3.2% rms in r, against 1.0–1.8% (median 1.1%) for 12 similar nearby stars. Too sparse to recover the period



Resolving the daily alias. A ground-based survey observes about once per night, so a true period appears alongside an alias offset by one sidereal day. ASAS-SN's two highest peaks, 1.3848 d and 3.5643 d, are such a pair: 1/1.3848 + 1/3.5643 = 1.0027 cycles per day. TESS observes continuously, and it has zero power at 3.5643 d in all five sectors. So 1.385 d is the true period. ATLAS's Fourier "dominant period" of 3.5635 d is the same alias; its Lomb–Scargle period, 2.769157 d, is twice the true period.

Period agreement: ASAS-SN 1.3848 d, ATLAS 1.38458 d and TESS 1.3849 d agree to within 0.03%, across observations from 2014 to 2023.

Gaia's proxy is the scatter implied by the uncertainty of its mean magnitude, $\sqrt{N}$ divided by `phot_g_mean_flux_over_error`. It isn't a light curve, but a high value marks a star as variable.

Catalogue status

No variable-star catalogue lists this star as a confirmed variable. It appears in three catalogues, none of which classifies it as a confirmed variable or gives the 1.385-day period.

Catalogue	Listed?	Entry
ATLAS variables (Heinze et al. 2018)	Yes, as a candidate	ATO J093.6734+17.5725, class `dubious` (probability 0.79), irregular 0.21
ROSAT Galactic plane survey (2001MNRAS.326.1161M)	Yes	Optical counterpart "13A" of X-ray source [MBP2001] 13, about 12″ away
Kounkel & Covey 2019, young groups (2019AJ....158..122K)	Yes	Member of Theia 399: 58 stars, log age 8.13 (about 135 Myr)
SIMBAD	Yes, not as variable	2MASS J06144163+1734211, type `*`
VSX	No	—
ASAS-SN variable stars	No	—
ZTF periodic variables (Chen et al. 2020)	No	—
Gaia DR3 variability (classifier, rotation, eclipsing, short-timescale)	No	`phot_variable_flag` not available
WISE variables	No	—
ExoFOP-TESS	No notes	Checked by hand

Variability catalogues were searched within 15″ and SIMBAD within 20″.

Theia 399 membership is uncertain. The group's mean parallax is 1.478 mas (about 677 pc) against the star's 1.667 mas, while the group is only about 10 pc across. The group's mean radial velocity is 0.8 km/s against the star's 24.5 km/s.

Nature of the variability

The evidence points to starspots on a young, magnetically active star, most likely in an unresolved binary where both stars are spotted.

Explanation	For	Against	Verdict
Spotted star(s) rotating at 1.385 d (BY Dra type)	Amplitude changed 2.5× between 2021 and 2023; shape and minima vary between sectors; X-ray source; possibly about 135 Myr old	A single star can't easily explain a second, stable 2.28 d period	Favoured, as part of a binary
Unresolved binary, both stars spotted (rotation periods 1.385 d and 2.28 d)	Two independent periods from the same position; Gaia RUWE 1.67; Gaia RV offset from the group	Unconfirmed until spectroscopy shows two sets of lines or a changing velocity	Most likely
Close tidally locked spotted binary (RS CVn type)	X-ray emission; RUWE	A locked binary would show one period, not two unrelated ones	Less likely
Ellipsoidal or eclipsing binary at 1.385 d or 2.77 d	Double-humped ATLAS fold	Ellipsoidal amplitude is fixed by geometry, but this one changed 2.5×; no eclipses seen	Ruled out
Pulsation	—	1–2 day periods with changing amplitude don't fit pulsating G stars	Ruled out

- **Differential rotation** could in principle give two periods. A 65% difference between them is much larger than typical for Sun-like stars, though.
- **Radius:** two similar stars in one unresolved image would make the system look about 0.75 mag brighter than either star. That inflates the radius estimated from brightness. The catalogue radius of 1.48 $R_\odot$ would then correspond to two stars of about 1.05 $R_\odot$ each. This fits the binary picture, but it isn't a measurement.

- **VSX type:** `BY` (BY Draconis, spotted dwarf) is the submitted type. `RS` (RS CVn) would apply only if the companion is found to be close and tidally locked.

Caveats and open questions

The period is secure; the star's nature is not yet. Spectroscopy is the decisive next step.

- **Blending in the surveys:** TESS pixels are 21″ and ASAS-SN apertures are about 16″ in radius. Both include some light from TIC 434331772, 30″ away. The TESS pixel maps show that star isn't the source, but survey amplitudes may be slightly diluted.
- **Sector 43:** the 2.28-day period dominates there, with the 1.385-day signal weak. A spot pattern that nearly cancels is a plausible cause, but it isn't proven.
- **The 2.28-day period:** its significance comes from standard false-alarm probabilities, which assume uncorrelated noise. Spot variability is correlated, so these overstate it. Its consistent period across four sectors is the stronger evidence.
- **Binary hypothesis:** RUWE 1.67, the two periods and the velocity offset are all indirect. Radial velocities or a spectrum showing two sets of lines would test it.
- **Youth:** membership of Theia 399 would make the star about 135 Myr old, but the parallax and radial-velocity tensions leave membership uncertain. Lithium absorption at 6708 Å in a spectrum would test youth directly.
- **X-ray association:** the ROSAT position lies about 12″ away, within that survey's typical errors. The identification comes from the 2001 survey paper and was not re-checked here.
- **Amplitude in 2026:** the January 2026 data hint at a lower amplitude, but 7 days is too short to say.

Data and methods

All analysis used public data and the Planet Finder scripts (Python 3.14, astropy 8, lightkurve 2.6).

Data	Details	Access
TESS light curves	QLP, sectors 43–45 (600 s) and 71–72 (200 s)	MAST via lightkurve

Data	Details	Access
TESS images	11×11-pixel full-frame-image cutouts, sectors 43–45, 71, 72 and the January 2026 "1751" segment	TESScut via lightkurve
ASAS-SN	Sky Patrol light curve AP24683508, camera ba, V band	ASAS-SN Sky Patrol, downloaded by hand
ATLAS	Variable-star catalogue, Heinze et al. 2018	VizieR J/AJ/156/241
ZTF	Light curves, objects 1558215400012041 (r) and 1558115400006285 (g), catflags = 0	IRSA ZTF light-curve service
Gaia DR3	gaia_source and variability tables	Gaia archive, VizieR I/358
Young groups	Kounkel & Covey 2019	VizieR J/AJ/158/122
Identifications	SIMBAD, including the ROSAT survey 2001MNRAS.326.1161M	SIMBAD

- **Periods:** Lomb–Scargle periodograms (astropy). For TESS, 1 day was trimmed from each data segment's ends. False-alarm probabilities use astropy's baluev method.
- **Second period:** a two-harmonic fit at 1.3849 d plus a linear trend was subtracted per sector, then the periodogram was repeated.
- **Pixel maps:** each cutout frame was sky-subtracted and binned to 30 minutes. Trends slower than 3 days were removed per pixel, then a two-harmonic wave was fitted to each pixel.
- **Comparison stars:** the Gaia comparison used all stars with G = 12.85–13.85 within 30′. The ZTF comparison used 12 Gaia stars with G = 13.1–13.6 within 15′.